

SDAIA ACADEMY

أكاديمية سدايا

Modular Stackable Learning Catalog

كتالوج التعلم المعياري القابل للتكديس

AI Engineer & Data Scientist Professional Pathways

31 stackable modules · 3 levels · 580 learning hours

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1. Catalog Design Methodology

This catalog redesigns SDAIA Academy's AI Engineer and Data Scientist learning paths into a modular, stackable architecture. It consolidates the Academy's prior program catalog, the competency traceability design, and two existing specialist programmes (Advanced Agentic AI Systems and Modern Data Engineering) into a single coherent offering aligned with the national workforce classification.

Design principles

- Backward design — every module starts from measurable, competency-focused learning outcomes aligned with revised Bloom's Taxonomy; content and assessment are derived from them.
- Competency-based progression — three levels (Practitioner → Specialist → Expert) mirror SDAIA's national classification of target audiences (مبتدئ / متخصص / خبير).
- Modular and stackable — each module is 2–5 days (5 learning hours per day), functions standalone, and issues a digital badge that stacks into certificates and specializations.
- Adult learning and project-based practice — every technical module dedicates at least 40% of time to labs and culminates in an applied artefact; both tracks end with panel-assessed capstones.
- Industry alignment — content reflects current practice: generative AI engineering, RAG, agentic systems, MLOps/LLMOps, Lakehouse data engineering, causal inference, and AI governance under PDPL and NDMO frameworks.
- Scalability — the coding scheme SDA-[TRACK]-[LEVEL][SEQ] accommodates future tracks (e.g., AI Product Manager, AI Solution Architect) without restructuring.

What changed from the previous catalog

- Durations standardised to 2–5 training days (10–25 hours) so modules are schedulable and stackable; multi-week programmes were decomposed.
- Gaps closed: prompt engineering foundations, LLM application engineering, fine-tuning/PEFT, AI security and red teaming, LLMOps observability, streaming analytics, decision science, and solution architecture.
- Redundancy removed: a single shared Foundation Core replaces duplicated intro content; Applied ML Foundations is cross-listed to both tracks instead of taught twice.
- Sequencing corrected: every module now declares explicit prerequisites and what it unlocks; security and governance appear before production deployment.
- The two existing Arabic programmes were absorbed as flagship Expert/Specialist modules (SDA-AIE-311, SDA-DSC-214) with their objectives rewritten as measurable outcomes.

2. Levels and Audience Classification

Levels follow the SDAIA Academy target-audience classification methodology and NSDAI workforce targets.

Practitioner / ممارس	Entry tier. Builds core technical and functional fluency. Open to professionals and graduates; low technical baseline assumed. Exit: contributes to AI/data work under guidance.
Specialist / متخصص	Mid tier. Intermediate technical skills applied to real delivery. Exit: independently delivers models, pipelines, and GenAI applications.
Expert / خبير	Top tier. Advanced practitioners who architect, secure, optimise, and operate production and national-scale systems, and lead teams. Exit: panel-defended capstone.

3. Learning Tracks

Foundation Core / النواة الأساسية	Shared on-ramp for all learners: AI and data literacy, generative-AI fluency, Python, data wrangling and SQL, and responsible AI under Saudi governance frameworks. Feeds both role tracks.
AI Engineer / مهندس الذكاء الاصطناعي	Develops engineers who design, build, secure, and operate production AI systems — from applied ML and deep learning through LLM applications, RAG, agentic systems, MLOps/LLMOps, security, and architecture.
Data Scientist / عالم البيانات	Develops scientists who turn data into decisions — statistics, advanced ML, forecasting, experimentation and causal inference, modern data engineering, streaming analytics, and decision science.

4. Stackable Certification Pathways

Each module issues a digital badge. Badges stack upward into level certificates, professional certificates, and cross-cutting micro-specializations. Full mapping is provided in the companion Excel catalog.

Level certificates	AI & Data Foundations (17 days) → Practitioner Certificate (29 days) → Specialist Certificate (45 days) → Expert Professional Certificate (AI Engineer 65 days / Data Scientist 59 days, incl. capstone).
Micro-specializations	GenAI Engineering (AIE-213/214/215/311) · MLOps & AI Reliability (AIE-216/312/314) · AI Security (FND-105 + AIE-313/311) · Analytics & Experimentation (DSC-212/213/311) · Data Engineering for AI (DSC-214/215/313).
Recognition of prior learning	Placement assessments allow experienced professionals to enter directly at Specialist or Expert level; equivalent portfolios may substitute capstone prerequisites.

5. Foundation Core Track Modules — النواة التأسيسية

SDA-FND-101 — AI and Data Literacy for Professionals

الإلمام بالذكاء الاصطناعي والبيانات للمهنيين

Level / المستوى	Practitioner / ممارس
Duration / المدة	2 days × 5 learning hours = 10 hours
Description	A shared on-ramp that builds a common language across AI, machine learning, generative AI, and the data value chain. Participants learn how modern AI systems create value, where they fail, and how data-driven decisions are made in organisations. The module anchors all subsequent pathways and requires no technical background.
الوصف	بوابة مشتركة تبني لغة موحدة حول الذكاء الاصطناعي وتعلم الآلة والذكاء الاصطناعي التوليدي وسلسلة قيمة البيانات. يتعرف المشاركون على كيفية تحقيق أنظمة الذكاء الاصطناعي للقيمة ومواطن إخفاقها. وكيفية اتخاذ القرارات المبينة على البيانات، دون الحاجة لخلفية تقنية.
Target Audience	Business professionals, decision-makers, fresh graduates, non-technical staff
Prerequisites	None — open entry point
Learning Outcomes	1. Explain core AI, ML, and generative-AI concepts and their business applications in plain language 2. Differentiate between rule-based systems, machine learning, deep learning, and generative AI 3. Analyze the data value chain from collection to decision within an organisational context 4. Evaluate AI use cases for feasibility, value, and risk using a structured framework 5. Identify common failure modes, biases, and limitations of AI systems 6. Interpret basic data visualisations and model outputs to support decisions
Topics	1. The AI, ML, and generative-AI landscape and terminology 2. How machine learning works: data, models, predictions 3. The data value chain and data-driven decision-making 4. Generative AI and large language models in the workplace 5. AI opportunity and risk assessment frameworks 6. National AI strategy context (NSDAI) and workforce roles 7. Hands-on: exploring AI tools and reading model outputs
Tools & Platforms	No-code AI tools, BI dashboards, AI assistants
Assessment	Quizzes; use-case evaluation exercise
Stackability	Entry badge; counts toward all certificates and both role tracks. Next: SDA-FND-102 / SDA-FND-103

SDA-FND-102 — Generative AI and Prompt Engineering Essentials

أساسيات الذكاء الاصطناعي التوليدي وهندسة الأوامر

Level / المستوى	Practitioner / ممارس
Duration / المدة	2 days × 5 learning hours = 10 hours
Description	A practical module on working productively and safely with generative-AI tools. Participants master prompt design patterns, output evaluation, and workflow integration for text, code, and data tasks. It equips every professional to use LLM-based assistants responsibly and effectively.
الوصف	وحدة تطبيقية للعمل بإنتاجية وأمان مع أدوات الذكاء الاصطناعي التوليدي. يتقن المشاركون أنماط تصميم الأوامر وتقييم المخرجات ودمجها في مهام النصوص والبرمجة والبيانات، بما يمكن كل مهني من استخدام المساعدات الذكية بمسؤولية وفعالية.
Target Audience	All professionals, analysts, technical and non-technical staff
Prerequisites	SDA-FND-101 (recommended)
Learning Outcomes	1. Explain how large language models generate output and why they hallucinate 2. Design effective prompts using role, context, format, and few-shot patterns 3. Apply iterative prompt refinement to improve output quality and reliability 4. Evaluate generative-AI outputs for accuracy, bias, and appropriateness 5. Integrate generative-AI assistants into document, analysis, and coding workflows 6. Assess data-privacy and confidentiality risks when using AI tools
Topics	1. How LLMs work: tokens, context, and limitations 2. Prompt design patterns and structured prompting 3. Iterative refinement and prompt evaluation 4. GenAI for documents, analysis, and code 5. Hallucination, bias, and output verification 6. Safe and compliant use of AI tools at work 7. Hands-on prompt engineering lab
Tools & Platforms	ChatGPT/Claude-class assistants, AI coding assistants, evaluation checklists
Assessment	Prompt-portfolio project; practical exercises
Stackability	Foundation badge; recommended before all GenAI engineering modules. Next: SDA-FND-103 or role-track modules

SDA-FND-103 — Python Programming for Data and AI

برمجة بايثون للبيانات والذكاء الاصطناعي

Level / المستوى	Practitioner / ممارس
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	An intensive hands-on module that builds job-ready Python skills for data and AI work. Participants progress from core language constructs to scientific computing with NumPy and pandas, version control, and clean, testable code. It is the technical gateway to both the AI Engineer and Data Scientist tracks.
الوصف	وحدة تطبيقية مكثفة تبني مهارات بايثون الجاهزة لسوق العمل في مجالي البيانات والذكاء الاصطناعي والتحكم Pandas و NumPy يتدرج المشاركون من أساسيات اللغة إلى الحوسبة العلمية باستخدام بالإصدارات وكتابة كود نظيف قابل للاختبار، وهي البوابة التقنية لمساري مهندس الذكاء الاصطناعي وعالم البيانات.
Target Audience	Aspiring AI engineers and data scientists, developers, technical graduates
Prerequisites	Basic computer literacy; programming logic recommended
Learning Outcomes	1. Develop structured Python programs using functions, data structures, and object-oriented design 2. Implement data manipulation and numerical computation workflows with pandas and NumPy 3. Apply version control with Git and collaborative repository workflows 4. Debug, test, and refactor Python code using systematic practices 5. Build reusable scripts and notebooks for data acquisition and processing 6. Organise projects with virtual environments and dependency management
Topics	1. Python core: types, control flow, functions, OOP 2. Data structures and file/API data handling 3. NumPy arrays and vectorised computation 4. pandas for tabular data manipulation 5. Git, GitHub, and collaboration workflows 6. Testing, debugging, and code quality 7. Environments and dependency management 8. Mini-project: end-to-end data processing pipeline
Tools & Platforms	Python, NumPy, pandas, Git/GitHub, VS Code, pytest, Jupyter
Assessment	Daily labs; coding assignments; mini-project
Stackability	Core technical badge; prerequisite for all track-level technical modules. Next: SDA-FND-104

SDA-FND-104 — Data Wrangling, SQL and Exploratory Analysis

والتحليل الاستكشافي SQL معالجة البيانات و

Level / المستوى	Practitioner / ممارس
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	A practical module covering the full journey from raw data to analysis-ready insight. Participants query relational databases with SQL, clean and reshape messy datasets, engineer features, and conduct rigorous exploratory data analysis. The module instils the data-quality discipline every AI and analytics role depends on.
الوصف	وحدة عملية تغطي الرحلة الكاملة من البيانات الخام إلى الرؤى الجاهزة للتحليل. يستعلم المشاركون من وينظفون البيانات ويعيدون تشكيلها، وبينون الخصائص، قواعد البيانات العلائقية باستخدام وينفذون تحليلًا استكشافيًا دقيقًا يرسخ انضباط جودة البيانات الذي تعتمد عليه كل أدوار الذكاء الاصطناعي والتحليلات.
Target Audience	Analysts, engineers, and scientists preparing for track-level modules
Prerequisites	SDA-FND-103 (Python)
Learning Outcomes	1. Design and execute SQL queries including joins, aggregations, and window functions 2. Diagnose data-quality issues and implement systematic cleaning strategies 3. Transform and reshape datasets for analysis using pandas 4. Engineer meaningful features from raw structured data 5. Conduct exploratory data analysis to surface patterns, anomalies, and relationships 6. Communicate EDA findings through clear visual and written summaries
Topics	1. Relational databases and SQL fundamentals 2. Advanced SQL: joins, aggregation, window functions 3. Data-quality assessment and cleaning workflows 4. Reshaping, merging, and transforming datasets 5. Feature engineering for analytics and ML 6. Exploratory data analysis and visual profiling 7. EDA case study on a real government-style dataset
Tools & Platforms	SQL (PostgreSQL), pandas, Jupyter, Matplotlib/Seaborn
Assessment	SQL labs; EDA project with report
Stackability	Foundation badge; required for Data Scientist track, recommended for AI Engineer track. Next: SDA-FND-105 and role-track entry

SDA-FND-105 — Responsible AI and Data Governance

الذكاء الاصطناعي المسؤول وحوكمة البيانات

Level / المستوى	Practitioner / ممارس
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	A governance-focused module aligning practice with Saudi national frameworks including PDPL, NDMO data standards, and SDAIA AI-ethics principles. Participants learn to assess fairness, transparency, privacy, and accountability across the AI lifecycle. The module prepares professionals to build and operate AI systems that are compliant and trustworthy by design.
الوصف	وحدة متخصصة في الحوكمة توائم الممارسة مع الأطر الوطنية السعودية بما فيها نظام حماية البيانات الشخصية ومعايير مكتب إدارة البيانات الوطنية ومبادئ أخلاقيات الذكاء الاصطناعي الصادرة عن سدايا. يتعلم المشاركون تقييم العدالة والشفافية والخصوصية والمساءلة عبر دورة حياة الذكاء الاصطناعي لبناء أنظمة موثوقة ومتوافقة تصميمياً.
Target Audience	All AI and data practitioners, compliance and governance staff
Prerequisites	SDA-FND-101
Learning Outcomes	1. Explain Saudi data-protection requirements (PDPL) and NDMO governance standards 2. Apply SDAIA AI-ethics principles across the AI system lifecycle 3. Analyze AI systems for bias, fairness, and disparate impact 4. Evaluate privacy risks and apply data-minimisation and anonymisation techniques 5. Design documentation, model cards, and audit trails for accountable AI 6. Assess third-party AI services for compliance and risk
Topics	1. The responsible-AI landscape and SDAIA ethics principles 2. PDPL and NDMO data-governance frameworks 3. Bias, fairness, and impact assessment 4. Privacy-preserving data handling 5. Transparency, explainability, and documentation 6. AI risk management and compliance workflows 7. Case studies: governance in government AI deployments
Tools & Platforms	Model cards, bias-audit toolkits, PDPL/NDMO checklists
Assessment	Governance case analysis; compliance-assessment exercise
Stackability	Foundation badge; required for Expert-level certificates in both tracks. Next: Role-track modules

6. AI Engineer Track Modules — مهندس الذكاء الاصطناعي

SDA-AIE-111 — Applied Machine Learning Foundations

أسس تعلم الآلة التطبيقي

Level / المستوى	Practitioner / ممارس
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	A hands-on foundation in supervised and unsupervised learning that teaches the complete modelling workflow. Participants frame business problems as ML tasks, train and tune models with scikit-learn, and evaluate them with the right metrics. The module ends with an end-to-end modelling project on real data.
الوصف	أساس تطبيقي في التعلم الخاضع وغير الخاضع للإشراف يغطي سير عمل النمذجة الكامل. يصوغ ويقيّمونها scikit-learn المشاركون المشكلات كمهام تعلم آلة ويدربون النماذج ويضبطونها باستخدام بالمقاييس المناسبة، وتختتم الوحدة بمشروع نمذجة متكامل على بيانات حقيقية.
Target Audience	Developers, engineers, analysts entering AI engineering; cross-listed to Data Scientist track
Prerequisites	SDA-FND-103; SDA-FND-104 recommended
Learning Outcomes	1. Frame business problems as supervised or unsupervised learning tasks 2. Implement the end-to-end ML workflow from data splitting to model selection 3. Develop classification and regression models using scikit-learn pipelines 4. Apply feature engineering, scaling, and encoding techniques appropriately 5. Evaluate models using suitable metrics, cross-validation, and error analysis 6. Diagnose and mitigate overfitting, underfitting, and data leakage 7. Optimize model performance through systematic hyperparameter tuning
Topics	1. ML problem framing and workflow design 2. Supervised learning: regression and classification 3. Feature engineering and preprocessing pipelines 4. Model evaluation, metrics, and cross-validation 5. Tree ensembles: random forests and gradient boosting 6. Unsupervised learning: clustering and dimensionality reduction 7. Hyperparameter tuning and model selection 8. End-to-end modelling project
Tools & Platforms	scikit-learn, pandas, Jupyter, XGBoost, Matplotlib
Assessment	Daily labs; end-to-end ML project
Stackability	Core ML badge; required for all Specialist AI Engineer modules and DSC-211. Next: SDA-AIE-112

SDA-AIE-112 — Deep Learning Essentials with PyTorch

PyTorch أساسيات التعلم العميق باستخدام

Level / المستوى	Practitioner / ممارس
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A practical introduction to neural networks that takes engineers from backpropagation mechanics to training real models on GPUs. Participants build, train, and debug networks in PyTorch, apply transfer learning, and manage experiments professionally. It is the gateway to NLP, computer vision, and generative-AI specialisations.
الوصف	مدخل عملي للشبكات العصبية ينقل المهندسين من آليات الانتشار الخلفي إلى تدريب نماذج حقيقية على مع تطبيق PyTorch وحدات معالجة الرسومات. يبني المشاركون الشبكات ويدربونها ويصححونها في التعلم بالنقل وإدارة التجارب باحترافية، وهي بوابة تخصصات معالجة اللغة والرؤية الحاسوبية والذكاء التوليدي.
Target Audience	Engineers who completed Applied ML; developers moving into deep learning
Prerequisites	SDA-AIE-111
Learning Outcomes	1. Explain neural-network architectures, activation functions, and backpropagation 2. Implement feed-forward, convolutional, and recurrent networks in PyTorch 3. Develop robust training loops with proper loss functions and optimisers 4. Apply regularisation, normalisation, and learning-rate scheduling to stabilise training 5. Implement transfer learning with pretrained models for new tasks 6. Track, compare, and reproduce experiments using experiment-management tools
Topics	1. Neural networks, activations, and backpropagation 2. PyTorch tensors, autograd, and training loops 3. CNNs for images; RNNs and sequence basics 4. Regularisation, normalisation, and optimisation 5. Transfer learning and pretrained models 6. GPU training and experiment tracking 7. Lab: build and tune an image classifier
Tools & Platforms	PyTorch, torchvision, Weights & Biases, CUDA, Hugging Face Hub
Assessment	Labs; model-training project
Stackability	Deep-learning badge; gateway to all Specialist AI Engineer modules. Next: SDA-AIE-113 / SDA-AIE-211

SDA-AIE-113 — Software Engineering Practices for AI Systems

ممارسات هندسة البرمجيات لأنظمة الذكاء الاصطناعي

Level / المستوى	Practitioner / ممارس
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	A production-engineering module that turns model builders into software engineers. Participants design clean APIs around models, containerise services with Docker, automate testing and CI/CD, and structure AI codebases for maintainability. These practices underpin every deployment module later in the track.
الوصف	وحدة هندسة إنتاجية تحوّل بناء النماذج إلى مهندسي برمجيات. يصمم المشاركون واجهات برمجية نظيفة ويؤتمتون الاختبارات وخطوط Docker حول النماذج ويحولون الخدمات إلى حاويات باستخدام CI/CD. وينظمون قواعد الكود لقابلية الصيانة، وهي ممارسات تؤسس لكل وحدات النشر اللاحقة.
Target Audience	AI practitioners preparing for production engineering roles
Prerequisites	SDA-FND-103; SDA-AIE-111 recommended
Learning Outcomes	1. Design RESTful APIs that expose ML models as reliable services 2. Implement containerised AI services using Docker and compose workflows 3. Develop automated unit, integration, and model-behaviour tests 4. Build CI/CD pipelines that validate and ship AI code changes 5. Apply clean-architecture and configuration-management patterns to AI codebases 6. Evaluate code quality using reviews, linting, and static analysis
Topics	1. Structuring AI projects and clean architecture 2. APIs for model serving with FastAPI 3. Docker and containerised services 4. Testing strategies for ML code 5. CI/CD automation with GitHub Actions 6. Configuration, secrets, and logging practices
Tools & Platforms	FastAPI, Docker, pytest, GitHub Actions, pydantic
Assessment	Labs; containerised model-service project
Stackability	Engineering badge; required for MLOps and production modules. Next: SDA-AIE-211 / SDA-AIE-216

SDA-AIE-211 — Natural Language Processing with Transformers

معالجة اللغة الطبيعية باستخدام المحولات

Level / المستوى	Specialist / متخصص
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A specialist module on modern NLP built around the transformer architecture. Participants tokenise and model text, fine-tune pretrained transformers for classification and extraction tasks, and handle Arabic-language processing challenges. The module bridges classic NLP pipelines and today's LLM-centric stack.
الوصف	وحدة تخصصية في معالجة اللغة الطبيعية الحديثة مبنية حول معمارية المحولات. يتعامل المشاركون مع الترميز ونمذجة النصوص وضبط المحولات المدربة مسبقًا لمهام التصنيف والاستخراج مع معالجة تحديات اللغة العربية، وتجسر الوحدة بين خطوط المعالجة الكلاسيكية ومنظومة النماذج اللغوية الحديثة.
Target Audience	AI engineers specialising in language applications
Prerequisites	SDA-AIE-112
Learning Outcomes	1. Implement text-preprocessing and tokenisation pipelines including Arabic-specific handling 2. Explain attention mechanisms and the transformer architecture 3. Fine-tune pretrained transformer models for classification, NER, and QA tasks 4. Develop embedding-based semantic search and similarity applications 5. Evaluate NLP models with task-appropriate metrics and error analysis 6. Optimize inference speed and memory for transformer deployment
Topics	1. Text processing, tokenisation, and embeddings 2. Attention and the transformer architecture 3. Fine-tuning BERT-family models for downstream tasks 4. Arabic NLP: challenges, models, and resources 5. Semantic search with sentence embeddings 6. Evaluation and error analysis for NLP 7. Inference optimisation and serving
Tools & Platforms	Hugging Face Transformers, spaCy, sentence-transformers, CAMEL Tools
Assessment	Labs; Arabic/English NLP application project
Stackability	NLP specialisation badge; feeds the GenAI Engineering specialisation. Next: SDA-AIE-213

SDA-AIE-212 — Computer Vision Systems Development

تطوير أنظمة الرؤية الحاسوبية

Level / المستوى	Specialist / متخصص
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A specialist module for building production computer-vision capabilities. Participants develop image classification, object detection, and segmentation solutions using modern architectures and transfer learning. The module emphasises dataset curation, augmentation, and deployment of vision models to real applications.
الوصف	وحدة تخصصية لبناء قدرات الرؤية الحاسوبية الإنتاجية. يطور المشاركون حلول تصنيف الصور واكتشاف الكائنات والتجزئة باستخدام المعماريات الحديثة والتعلم بالنقل، مع تركيز على تنسيق مجموعات البيانات والتعزيز ونشر نماذج الرؤية في تطبيقات حقيقية.
Target Audience	AI engineers specialising in vision applications
Prerequisites	SDA-AIE-112
Learning Outcomes	1. Implement image classification pipelines using CNNs and vision transformers 2. Develop object-detection and segmentation solutions with modern frameworks 3. Design dataset curation, labelling, and augmentation strategies 4. Apply transfer learning and fine-tuning to domain-specific vision tasks 5. Evaluate vision models using appropriate metrics such as mAP and IoU 6. Deploy optimised vision models to cloud and edge targets
Topics	1. Image data, preprocessing, and augmentation 2. CNN architectures and vision transformers 3. Object detection with YOLO-family models 4. Semantic and instance segmentation 5. Dataset curation and annotation workflows 6. Model evaluation for vision tasks 7. Deployment and edge optimisation 8. Vision application project
Tools & Platforms	PyTorch, torchvision, YOLO/Ultralytics, OpenCV, ONNX
Assessment	Labs; vision-system project
Stackability	Vision specialisation badge; elective within AI Engineer Specialist certificate. Next: SDA-AIE-216

SDA-AIE-213 — Large Language Model Application Engineering

هندسة تطبيقات النماذج اللغوية الكبيرة

Level / المستوى	Specialist / متخصص
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A hands-on module for engineering applications on top of LLM APIs and open-weight models. Participants design prompt pipelines, structured outputs, function calling, and evaluation harnesses, and architect complete LLM-powered products. It establishes the engineering discipline behind reliable generative-AI software.
الوصف	وحدة تطبيقية لهندسة التطبيقات فوق واجهات النماذج اللغوية الكبيرة والنماذج مفتوحة الأوزان. يصمم المشاركون خطوط الأوامر والمخرجات المهيكلية واستدعاء الوظائف ومنصات التقييم ويبنون منتجات متكاملة مدعومة بالنماذج اللغوية، بما يرسخ الانضباط الهندسي وراء برمجيات الذكاء التوليدي الموثوقة.
Target Audience	Engineers building generative-AI products and services
Prerequisites	SDA-AIE-111; SDA-FND-102 recommended
Learning Outcomes	1. Design application architectures around LLM APIs and open-weight models 2. Implement structured outputs, function calling, and tool integration 3. Develop robust prompt pipelines with templating and guardrails 4. Build evaluation harnesses to measure quality, safety, and regression 5. Optimize latency and token cost across model and caching choices 6. Compare commercial and open-weight models for a given use case
Topics	1. LLM application architecture patterns 2. Working with LLM APIs and open-weight models 3. Structured outputs and function calling 4. Prompt pipelines, templating, and guardrails 5. LLM evaluation: golden sets, LLM-as-judge, metrics 6. Cost, latency, and caching strategies 7. Building a complete LLM-powered application
Tools & Platforms	OpenAI/Anthropic APIs, LangChain, pydantic, promptfoo/evals, vLLM
Assessment	Labs; LLM application project with evaluation report
Stackability	GenAI engineering badge; core of the GenAI Engineering specialisation. Next: SDA-AIE-214

SDA-AIE-214 — Building Retrieval-Augmented Generation Systems

بناء أنظمة التوليد المعزز بالاسترجاع

Level / المستوى	Specialist / متخصص
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A deep, practical module on grounding LLMs in enterprise knowledge. Participants build the full RAG pipeline — ingestion, chunking, embedding, vector retrieval, reranking, and generation — and evaluate answer faithfulness systematically. The module covers advanced patterns including hybrid search and multi-document reasoning for production-grade assistants.
الوصف	الكامل من RAG وحدة عملية متعمقة في تأصيل النماذج اللغوية بمعرفة المؤسسة. يبني المشاركون خط الاستيعاب والتقطيع والتضمين والاسترجاع المتجهي وإعادة الترتيب إلى التوليد، مع تقييم منهجي لأمانة الإجابات، وتغطي الوحدة أنماطًا متقدمة تشمل البحث الهجين والاستدلال متعدد المستندات للمساعدات الإنتاجية.
Target Audience	GenAI engineers building knowledge assistants and enterprise search
Prerequisites	SDA-AIE-213
Learning Outcomes	1. Design end-to-end RAG architectures for enterprise knowledge bases 2. Implement ingestion, chunking, and embedding pipelines for heterogeneous documents 3. Develop retrieval stacks combining vector, keyword, and hybrid search with reranking 4. Evaluate RAG quality using faithfulness, relevance, and answer-correctness metrics 5. Optimize retrieval parameters, context construction, and citation grounding 6. Validate systems against hallucination and stale-knowledge failure modes
Topics	1. RAG architecture and when to use it versus fine-tuning 2. Document ingestion, parsing, and chunking strategies 3. Embeddings and vector databases 4. Hybrid retrieval and reranking 5. Context construction and citation grounding 6. RAG evaluation frameworks 7. Advanced patterns: multi-hop, agentic RAG 8. Production RAG project
Tools & Platforms	Vector DBs (Qdrant/pgvector), LangChain/LlamaIndex, RAGAS, rerankers
Assessment	Labs; enterprise RAG project with evaluation
Stackability	RAG badge; prerequisite for Agentic AI Systems Engineering. Next: SDA-AIE-311

SDA-AIE-215 — Fine-Tuning and Adapting Foundation Models

الضبط الدقيق وتكييف النماذج الأساسية

Level / المستوى	Specialist / متخصص
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	A specialist module on customising foundation models efficiently. Participants prepare instruction datasets, apply parameter-efficient fine-tuning (LoRA/QLoRA), align models with preference optimisation, and quantise them for deployment. The module gives engineers a decision framework for when to prompt, retrieve, or fine-tune.
الوصف	وحدة تخصصية في تخصيص النماذج الأساسية بكفاءة. يجهز المشاركون بيانات التعليمات ويطبقون ومواءمة النماذج بتحسين التفضيلات وتكميمها (LoRA/QLoRA) الضبط الدقيق الموفر للمعاملات للنشر، وتمنح الوحدة المهندسين إطار قرار للاختيار بين الأوامر أو الاسترجاع أو الضبط الدقيق.
Target Audience	GenAI engineers customising models for domain tasks
Prerequisites	SDA-AIE-112; SDA-AIE-213 recommended
Learning Outcomes	1. Compare prompting, RAG, and fine-tuning strategies for a given requirement 2. Design high-quality instruction and preference datasets 3. Implement parameter-efficient fine-tuning using LoRA and QLoRA 4. Apply quantisation techniques to reduce model footprint 5. Evaluate fine-tuned models against baselines with rigorous benchmarks 6. Validate adapted models for regression, safety, and domain fitness
Topics	1. Adaptation strategy: prompt vs RAG vs fine-tune 2. Instruction dataset design and curation 3. PEFT methods: LoRA, QLoRA, adapters 4. Preference alignment (DPO) overview 5. Quantisation and efficient inference 6. Evaluation and regression testing of tuned models
Tools & Platforms	Hugging Face PEFT/TRL, bitsandbytes, Axolotl, lm-eval-harness
Assessment	Labs; domain fine-tuning project
Stackability	Model-adaptation badge; elective within GenAI Engineering specialisation. Next: SDA-AIE-311 / SDA-AIE-312

SDA-AIE-216 — MLOps: Model Deployment and Automation

عمليات تعلم الآلة: نشر النماذج والأتمتة

Level / المستوى	Specialist / متخصص
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	A comprehensive module on operationalising machine learning. Participants build reproducible training pipelines, register and version models and data, deploy scalable serving endpoints, and automate retraining with monitoring for drift. The module establishes the MLOps backbone that Expert-level reliability and LLMOps modules extend.
الوصف	وحدة شاملة في تشغيل تعلم الآلة. يبني المشاركون خطوط تدريب قابلة للتكرار ويسجلون النماذج والبيانات بإصدارات ويطلقون نقاط خدمة قابلة للتوسع ويؤتمتون إعادة التدريب مع مراقبة الانحراف، بما للخبراء LLMOps يؤسس العمود الفقري لعمليات تعلم الآلة الذي تبني عليه وحدات الموثوقية و
Target Audience	AI engineers responsible for production ML systems
Prerequisites	SDA-AIE-113; SDA-AIE-111
Learning Outcomes	1. Design end-to-end MLOps architectures spanning training, serving, and monitoring 2. Implement reproducible pipelines with data and model versioning 3. Deploy models as scalable, containerised inference services 4. Develop automated retraining and CI/CD/CT workflows 5. Implement monitoring for data drift, concept drift, and performance decay 6. Evaluate deployment strategies including canary, blue-green, and shadow releases
Topics	1. MLOps maturity model and system architecture 2. Experiment tracking and model registry 3. Data versioning and feature stores 4. Pipeline orchestration and automation 5. Model serving: batch, online, streaming 6. Drift detection and monitoring 7. Release strategies and rollback 8. MLOps platform project
Tools & Platforms	MLflow, DVC, Airflow/Prefect, Docker, Kubernetes basics, Evidently
Assessment	Labs; deployed pipeline project
Stackability	MLOps badge; prerequisite for LLMOps and reliability modules. Next: SDA-AIE-312

SDA-AIE-311 — Agentic AI Systems Engineering

هندسة أنظمة الذكاء الاصطناعي الوكيل

Level / المستوى	Expert / خبير
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	An expert module on designing, building, securing, and operating enterprise agentic systems that plan, reason, use tools, and manage memory across multi-step workflows. Participants implement ReAct, Plan-and-Execute, and Reflection patterns, orchestrate multi-agent architectures, integrate tools via the Model Context Protocol, and harden agents with guardrails and red-team testing. The module culminates in deploying a production-grade agentic system in a simulated enterprise environment.
الوصف	وحدة خبراء في تصميم وبناء وتأمين وتشغيل أنظمة وكيلية مؤسسية قادرة على التخطيط والاستدلال و ReAct واستخدام الأدوات وإدارة الذاكرة عبر سير عمل متعدد الخطوات. يطبق المشاركون أنماط وينسقون معماريات متعددة الوكلاء ويدمجون الأدوات عبر Reflection و Plan-and-Execute و MCP بروتوكول ويحصلون الوكلاء بالحوازر الوقائية واختبارات الفريق الأحمر، وتختتم الوحدة بنشر نظام وكيل بمستوى إنتاجي في بيئة مؤسسية محاكاة.
Target Audience	Senior AI engineers, AI architects, technical leads
Prerequisites	SDA-AIE-214
Learning Outcomes	1. Design agentic architectures with planning, reasoning, tool use, and memory 2. Implement ReAct, Plan-and-Execute, and Reflection patterns in working agents 3. Develop multi-agent systems with orchestration, delegation, and task routing 4. Integrate tools and enterprise APIs using the Model Context Protocol 5. Design state management and short/long-term memory with vector stores 6. Implement automated guardrails and red-team agents against attacks 7. Optimize agent performance, cost, and reliability with tracing and monitoring 8. Validate a deployed agentic system in a simulated enterprise environment
Topics	1. Agentic system architectures and components 2. Planning and reasoning patterns: ReAct, Plan-and-Execute, Reflection 3. Tool use, function calling, and MCP integration 4. Memory and state management with vector databases 5. Single-agent versus multi-agent architectures 6. Hierarchical delegation and agent orchestration 7. Flow engineering over prompt engineering 8. Guardrails, security, and adversarial testing 9. Observability, cost optimisation, and monitoring 10. Capstone: enterprise agentic system deployment
Tools & Platforms	LangGraph, MCP, vector DBs, agent-tracing platforms (LangSmith), guardrails frameworks
Assessment	Labs; security report; deployed agent capstone

Stackability

Flagship Expert badge; anchors the GenAI Engineering specialisation.
Derived from the existing Advanced Agentic AI Systems programme.
Next: SDA-AIE-313 / SDA-AIE-390

SDA-AIE-312 — LLMops and Production AI Observability

عمليات النماذج اللغوية ومراقبة الذكاء الاصطناعي الإنتاجي

Level / المستوى	Expert / خبير
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	An expert module on operating LLM and generative-AI workloads in production. Participants implement tracing, quality monitoring, feedback loops, and incident response for AI services, extending MLOps discipline to non-deterministic systems. The module covers evaluation-in-production, prompt/version management, and reliability engineering for AI platforms.
الوصف	وحدة خبراء في تشغيل أحمال النماذج اللغوية والذكاء التوليدي في الإنتاج. يطبق المشاركون التتبع ومراقبة الجودة وحلقات التغذية الراجعة والاستجابة للحوادث لخدمات الذكاء الاصطناعي، بما يمد إلى الأنظمة غير الحتمية، وتغطي الوحدة التقييم في الإنتاج وإدارة إصدارات الأوامر MLOps انضباط وهندسة الموثوقية لمنصات الذكاء الاصطناعي.
Target Audience	Engineers operating GenAI services at scale
Prerequisites	SDA-AIE-216; SDA-AIE-213
Learning Outcomes	1. Design observability architectures for LLM applications with distributed tracing 2. Implement online evaluation, quality scoring, and user-feedback loops 3. Develop prompt and model version-management workflows with safe rollouts 4. Analyze production incidents in AI systems and build response runbooks 5. Implement SLOs, alerting, and reliability dashboards for AI services 6. Optimize the operations lifecycle for continuous improvement of deployed LLM systems
Topics	1. From MLOps to LLMops: what changes 2. Tracing and logging for LLM pipelines 3. Online evaluation and quality monitoring 4. Prompt registries and version management 5. Feedback loops and data flywheels 6. SLOs, alerting, and incident response 7. Reliability engineering for AI platforms
Tools & Platforms	LangSmith/Langfuse, OpenTelemetry, Grafana, evaluation pipelines
Assessment	Labs; observability dashboard project
Stackability	Expert operations badge; part of the MLOps & Reliability specialisation. Next: SDA-AIE-314 / SDA-AIE-390

SDA-AIE-313 — AI Security, Guardrails and Red Teaming

أمن الذكاء الاصطناعي والحواجز الوقائية والفريق الأحمر

Level / المستوى	Expert / خبير
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	An expert module on securing AI systems against adversarial threats. Participants analyse attack surfaces including prompt injection, jailbreaks, data poisoning, and model extraction, then design layered defences with guardrails, filtering, and access controls. Structured red-team exercises teach systematic security validation of LLM and agentic applications.
الوصف	وحدة خبراء في تأمين أنظمة الذكاء الاصطناعي ضد التهديدات العدائية. يحلل المشاركون أسطح الهجوم بما فيها حقن الأوامر وكسر القيود وتسميم البيانات واستخراج النماذج، ثم يصممون دفاعات متعددة الطبقات بالحواجز الوقائية والتصفية وضوابط الوصول، مع تمارين فريق أحمر منظمة للتحقق الأمني المنهجي من تطبيقات النماذج اللغوية والوكيلية.
Target Audience	Senior engineers, security specialists, AI platform owners
Prerequisites	SDA-AIE-213; SDA-AIE-311 recommended
Learning Outcomes	1. Analyze AI attack surfaces using OWASP LLM Top-10 and threat modelling 2. Evaluate systems against prompt injection, jailbreak, and data-exfiltration attacks 3. Design layered guardrail architectures for inputs, outputs, and tools 4. Implement content filtering, PII protection, and policy enforcement 5. Conduct structured red-team exercises against LLM and agentic applications 6. Develop security test suites and reporting aligned with governance requirements
Topics	1. AI threat landscape and OWASP LLM Top-10 2. Prompt injection and jailbreak techniques 3. Data poisoning and supply-chain risks 4. Guardrail architecture and policy enforcement 5. Red-teaming methodology and tooling 6. Security evaluation and reporting 7. Compliance alignment with national frameworks
Tools & Platforms	Garak/PyRIT, guardrails frameworks, NeMo Guardrails, threat-modelling templates
Assessment	Red-team exercise; security assessment report
Stackability	Security badge; required for AI Engineer Expert certificate. Next: SDA-AIE-390

SDA-AIE-314 — AI Performance, Scalability and Cost Optimization

أداء الذكاء الاصطناعي وقابلية التوسع وتحسين التكلفة

Level / المستوى	Expert / خبير
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	An expert module on engineering AI systems that are fast, scalable, and economical. Participants profile inference workloads, apply batching, caching, quantisation, and routing strategies, and design autoscaling serving infrastructure. The module builds the FinOps discipline to run national-scale AI services within budget.
الوصف	وحدة خبراء في هندسة أنظمة ذكاء اصطناعي سريعة وقابلة للتوسع واقتصادية. يحلل المشاركون أحمال الاستدلال ويطبقون استراتيجيات التجميع والتخزين المؤقت والتكميم والتوجيه ويصممون بنية خدمة ذاتية لتشغيل خدمات ذكاء اصطناعي وطنية ضمن الميزانية FinOps التوسع، مع بناء انضباط.
Target Audience	Senior engineers and platform architects
Prerequisites	SDA-AIE-216
Learning Outcomes	1. Analyze latency, throughput, and cost profiles of AI inference workloads 2. Implement batching, caching, and streaming optimisations for model serving 3. Apply quantisation, distillation, and model-routing to cut serving costs 4. Design autoscaling GPU infrastructure for variable demand 5. Evaluate build-versus-buy and model-tier trade-offs with TCO analysis 6. Develop cost-monitoring dashboards and optimisation playbooks
Topics	1. Performance profiling of inference workloads 2. Serving optimisation: batching, caching, streaming 3. Model compression: quantisation and distillation 4. Model routing and cascade architectures 5. GPU infrastructure and autoscaling 6. AI FinOps: cost monitoring and TCO 7. Optimisation project on a live workload
Tools & Platforms	vLLM, TensorRT/ONNX Runtime, Kubernetes autoscaling, cost dashboards
Assessment	Optimisation project with before/after benchmark
Stackability	Optimisation badge; part of the MLOps & Reliability specialisation. Next: SDA-AIE-390

SDA-AIE-315 — Enterprise AI Solution Architecture

معمارية حلول الذكاء الاصطناعي المؤسسية

Level / المستوى	Expert / خبير
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	An expert module that elevates engineers to solution architects. Participants translate business requirements into end-to-end AI architectures, select platforms and integration patterns, and produce architecture documentation and decision records. Case-based design reviews cover real government and enterprise AI scenarios.
الوصف	وحدة خبراء ترتقي بالمهندسين إلى مستوى معماري الحلول. يحول المشاركون متطلبات الأعمال إلى معماريات ذكاء اصطناعي متكاملة ويختارون المنصات وأنماط التكامل وينتجون وثائق المعمارية وسجلات القرارات، مع مراجعات تصميم قائمة على حالات حكومية ومؤسسية حقيقية.
Target Audience	Senior engineers moving into architecture and technical leadership
Prerequisites	SDA-AIE-216; SDA-AIE-213
Learning Outcomes	1. Translate business requirements into candidate AI solution architectures 2. Design integration patterns connecting AI services to enterprise systems 3. Evaluate platform, cloud, and sovereignty options against constraints 4. Design for security, compliance, and data residency from the outset 5. Develop architecture decision records and reference documentation 6. Defend architectural choices in structured design reviews
Topics	1. AI solution architecture lifecycle 2. Reference architectures: RAG, agents, ML platforms 3. Enterprise integration and API strategy 4. Cloud, on-premise, and sovereignty considerations 5. Non-functional requirements: security, scale, compliance 6. Architecture documentation and ADRs 7. Design-review simulations on real scenarios
Tools & Platforms	Architecture canvases, C4 modelling, cloud reference architectures
Assessment	Architecture design package; peer design review
Stackability	Architecture badge; elective for Expert certificate; on-ramp to future AI Solution Architect track. Next: SDA-AIE-390

SDA-AIE-390 — AI Engineering Capstone: Production GenAI System

مشروع التخرج: نظام ذكاء توليدي إنتاجي

Level / المستوى	Expert / خبير
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	The integrative capstone of the AI Engineer pathway. Teams scope, design, build, secure, and deploy a complete generative-AI system — combining RAG or agents, MLOps automation, observability, and guardrails — for a realistic enterprise brief. Work is assessed through a production demo, security review, and architecture defence before an expert panel.
الوصف	المشروع التكاملي الختامي لمسار مهندس الذكاء الاصطناعي. تتولى الفرق تحديد نطاق نظام ذكاء توليدي والمراقبة MLOps أو الوكلاء وأتمتة RAG متكامل وتصميمه وبناءه وتأمينه ونشره — يجمع بين والحواجز الوقائية — وفق موجز مؤسسي واقعي، ويُقيّم العمل عبر عرض إنتاجي ومراجعة أمنية ودفاع معماري أمام لجنة خبراء.
Target Audience	Candidates completing the AI Engineer Expert certificate
Prerequisites	SDA-AIE-311 + SDA-AIE-312 + SDA-AIE-313 (or equivalent portfolio)
Learning Outcomes	1. Design a complete production architecture for an enterprise GenAI use case 2. Implement an integrated system combining retrieval, agents, and serving infrastructure 3. Develop automated evaluation, monitoring, and guardrail layers 4. Validate the system through security testing and load benchmarks 5. Optimize cost and performance against defined service objectives 6. Defend design decisions and results before a technical panel
Topics	1. Project scoping and requirements engineering 2. System design and architecture review 3. Sprint-based build with mentor checkpoints 4. Evaluation, security, and performance hardening 5. Deployment and operations readiness 6. Final demo, documentation, and panel defence
Tools & Platforms	Full track toolchain as selected by teams
Assessment	Deployed system; documentation package; panel defence
Stackability	Completes the AI Engineer Expert (Professional) Certificate.

7. Data Scientist Track Modules — عالم البيانات

SDA-DSC-111 — Statistical Foundations for Data Science

الأسس الإحصائية لعلم البيانات

Level / المستوى	Practitioner / ممارس
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A rigorous, applied grounding in the statistics that underpin trustworthy data science. Participants master probability, distributions, estimation, hypothesis testing, and regression, applying each concept computationally in Python. The module builds the inferential judgement needed to draw valid conclusions from data.
الوصف	تأسيس تطبيقي دقيق في الإحصاء الذي يقوم عليه علم البيانات الموثوق. يتقن المشاركون الاحتمالات والتوزيعات والتقدير واختبار الفرضيات والانحدار مع تطبيق كل مفهوم حاسوبيًا في بايثون، بما يبنى الحكم الاستدلالي اللازم لاستخلاص نتائج صحيحة من البيانات.
Target Audience	Analysts, graduates, and engineers entering data science
Prerequisites	SDA-FND-103; SDA-FND-104 recommended
Learning Outcomes	1. Apply probability rules and distributions to model real-world uncertainty 2. Analyze sampling variability using confidence intervals and standard errors 3. Implement hypothesis tests and interpret p-values and effect sizes correctly 4. Develop linear and logistic regression models and diagnose their assumptions 5. Evaluate statistical claims for common fallacies and misuse 6. Simulate statistical processes computationally to validate analytical results
Topics	1. Probability and common distributions 2. Descriptive statistics and sampling 3. Estimation and confidence intervals 4. Hypothesis testing and effect sizes 5. Correlation, causation, and confounding 6. Linear and logistic regression 7. Simulation and resampling methods
Tools & Platforms	Python, SciPy, statsmodels, Jupyter
Assessment	Labs; statistical analysis project
Stackability	Core statistics badge; required for all Specialist Data Scientist modules. Next: SDA-DSC-112 / SDA-AIE-111

SDA-DSC-112 — Data Visualization and Storytelling

تصور البيانات وسرد القصص

Level / المستوى	Practitioner / ممارس
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	A craft-focused module on turning analysis into influence. Participants apply perception science to design honest, compelling visualisations, build interactive dashboards, and structure data narratives for executive audiences. The module closes the last mile between analytical work and organisational decisions.
الوصف	وحدة مهارية لتحويل التحليل إلى تأثير. يطبق المشاركون علم الإدراك لتصميم تصورات صادقة ومقنعة وبناء لوحات معلومات تفاعلية وهيكلة سرديات البيانات لجمهور تنفيذي، بما يخلق المسافة الأخيرة بين العمل التحليلي وقرارات المؤسسة.
Target Audience	Data scientists, analysts, BI professionals
Prerequisites	SDA-FND-104
Learning Outcomes	1. Apply visual-perception principles to select effective chart types 2. Design clear, honest visualisations that avoid distortion and clutter 3. Develop interactive dashboards for self-service exploration 4. Construct data narratives structured around audience and decision 5. Evaluate visualisations through critique and iterative refinement 6. Present analytical findings persuasively to executive stakeholders
Topics	1. Perception science and visual encoding 2. Chart selection and design principles 3. Colour, annotation, and decluttering 4. Interactive dashboards and BI tools 5. Data storytelling structure and flow 6. Executive communication and presentation lab
Tools & Platforms	Plotly, Power BI/Tableau, Matplotlib/Seaborn
Assessment	Dashboard project; storytelling presentation
Stackability	Communication badge; required for Data Scientist Practitioner certificate. Next: SDA-DSC-211

SDA-DSC-211 — Advanced Machine Learning Methods

أساليب تعلم الآلة المتقدمة

Level / المستوى	Specialist / متخصص
Duration / المدة	4 days × 5 learning hours = 20 hours
Description	A specialist module that deepens modelling skill beyond the foundations. Participants master gradient-boosting frameworks, imbalanced and cost-sensitive learning, model interpretability with SHAP, and rigorous validation design. The emphasis is on winning reliable performance on messy, real-world tabular problems.
الوصف	وحدة تخصصية تعمق مهارة النمذجة إلى ما بعد الأساسيات. يتقن المشاركون أطر التعزيز المتدرج وتصميم تحقق SHAP والتعلم مع البيانات غير المتوازنة والحساسية للتكلفة وتفسير النماذج باستخدام صارم، مع تركيز على تحقيق أداء موثوق في مشكلات جدولية واقعية معقدة.
Target Audience	Data scientists progressing to specialist modelling roles
Prerequisites	SDA-AIE-111 (cross-listed); SDA-DSC-111
Learning Outcomes	1. Develop high-performance models with XGBoost and LightGBM 2. Design validation schemes that respect leakage, grouping, and time order 3. Apply techniques for imbalanced, noisy, and cost-sensitive problems 4. Analyze model behaviour using SHAP and permutation importance 5. Optimize feature pipelines and model ensembles for tabular data 6. Evaluate models for stability, calibration, and business impact
Topics	1. Gradient boosting deep dive 2. Validation design and leakage prevention 3. Imbalanced and cost-sensitive learning 4. Model interpretability: SHAP and beyond 5. Calibration and uncertainty estimates 6. Ensembling and stacking strategies 7. Applied modelling challenge
Tools & Platforms	XGBoost, LightGBM, SHAP, scikit-learn, Optuna
Assessment	Modelling challenge; interpretability report
Stackability	Advanced ML badge; core of the Data Scientist Specialist certificate. Next: SDA-DSC-212 / SDA-DSC-213

SDA-DSC-212 — Time Series Analysis and Forecasting

تحليل السلاسل الزمنية والتنبؤ

Level / المستوى	Specialist / متخصص
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	A specialist module on modelling data that evolves over time. Participants decompose series, build statistical and ML-based forecasting models, and evaluate them with time-aware backtesting. Applications include demand planning, workforce projections, and economic indicators relevant to national programmes.
الوصف	وحدة تخصصية في نمذجة البيانات المتغيرة عبر الزمن. يفكك المشاركون السلاسل ويبنّون نماذج تنبؤ إحصائية وقائمة على تعلم الآلة ويقيمونها باختبار رجعي مراعي للزمن، مع تطبيقات تشمل تخطيط الطلب وإسقاطات القوى العاملة والمؤشرات الاقتصادية ذات الصلة بالبرامج الوطنية.
Target Audience	Data scientists and analysts working with temporal data
Prerequisites	SDA-DSC-111; SDA-DSC-211 recommended
Learning Outcomes	1. Analyze trend, seasonality, and autocorrelation structure in time series 2. Develop classical forecasting models including ARIMA and exponential smoothing 3. Implement ML and gradient-boosting approaches with engineered temporal features 4. Design backtesting frameworks with proper time-based validation 5. Evaluate forecasts using scale-appropriate accuracy and uncertainty metrics 6. Compare model families to select the right approach per use case
Topics	1. Time-series structure and decomposition 2. Stationarity and autocorrelation 3. ARIMA and exponential smoothing 4. Feature-based ML forecasting 5. Probabilistic forecasts and intervals 6. Backtesting and forecast evaluation 7. Forecasting case study
Tools & Platforms	statsmodels, Prophet, sktime, LightGBM
Assessment	Forecasting project with backtest report
Stackability	Forecasting badge; part of Analytics & Experimentation specialisation. Next: SDA-DSC-311

SDA-DSC-213 — Experimentation, A/B Testing and Causal Inference

والاستدلال السببي A/B التجريب واختبارات

المستوى / Level	متخصص / Specialist
المدة / Duration	4 days × 5 learning hours = 20 hours
الوصف / Description	A specialist module on answering "what actually works" with data. Participants design and analyse randomised experiments, compute power and sample sizes, and apply causal-inference methods — matching, difference-in-differences, and instrumental variables — when experiments are impossible. The module turns data scientists into trusted advisors on policy and product decisions. وحدة تخصصية للإجابة عن سؤال "ما الذي يحدث الأثر فعلاً" بالبيانات. يصمم المشاركون التجارب العشوائية ويحللون نتائجها ويحسبون القوة وأحجام العينات ويطبقون أساليب الاستدلال السببي — المطابقة والفروق في الفروق والمتغيرات الأداة — حين تتعذر التجارب، بما يجعل عالم البيانات مستشاراً موثقاً لقرارات السياسات والمنتجات.
الوصف / الوصف	
الجمهور المستهدف / Target Audience	Data scientists supporting product, policy, and programme decisions
المتطلبات / Prerequisites	SDA-DSC-111
النتائج المتوقعة / Learning Outcomes	1. Design randomised experiments with valid randomisation units and guardrail metrics 2. Compute statistical power, sample sizes, and minimum detectable effects 3. Analyze experiment results with correct variance estimation and multiple-testing control 4. Apply quasi-experimental methods including matching and difference-in-differences 5. Evaluate causal claims from observational data using DAG-based reasoning 6. Communicate experimental evidence and uncertainty to decision-makers
المواضيع / Topics	1. Causal thinking and potential outcomes 2. Experiment design and randomisation 3. Power analysis and sample sizing 4. Analysis pitfalls: peeking, multiple testing 5. Quasi-experiments: matching, DiD, IV 6. Causal graphs and confounding 7. Experimentation case study
الأدوات والمنصات / Tools & Platforms	statsmodels, DoWhy/EconML, experiment simulators
التقييم / Assessment	Experiment design and analysis project
المتداخلة / Stackability	Causal badge; core of Analytics & Experimentation specialisation. Next: SDA-DSC-311

SDA-DSC-214 — Modern Data Engineering for AI Systems

هندسة البيانات الحديثة لأنظمة الذكاء الاصطناعي

Level / المستوى	Specialist / متخصص
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	A comprehensive module on building modern, scalable, production-ready data architectures that feed AI and analytics. Participants trace the evolution from warehouses and lakes to the Lakehouse, master Delta Lake with ACID transactions and compute-storage separation, and work with ELT, streaming, data quality, and governance. A hands-on build of a local Mini-Lakehouse with Delta Lake and PySpark anchors the learning.
الوصف	وحدة شاملة لبناء معماريات بيانات حديثة قابلة للتوسع وجاهزة للإنتاج تغذي الذكاء الاصطناعي ويتقنون Lakehouse والتحليلات. يتتبع المشاركون التطور من المستودعات والبحيرات إلى معمارية والتدفقات ELT وفصل الحوسبة عن التخزين ويعملون مع ACID مع معاملات Delta Lake محلي Mini-Lakehouse اللحظية وجودة البيانات والحوكمة، ويرسخ التعلم تطبيق عملي لبناء باستخدام Delta Lake وPySpark.
Target Audience	Data engineers, data scientists, systems and AI architects, data-infrastructure leads
Prerequisites	SDA-FND-104; SDA-FND-103
Learning Outcomes	1. Compare data warehouses, data lakes, and Lakehouse architectures across use cases 2. Explain compute-storage separation and its impact on cost, elasticity, and scale 3. Design ELT pipelines suited to modern cloud data environments 4. Implement Delta Lake tables with ACID transactions and schema enforcement 5. Develop streaming and event-driven ingestion patterns for real-time data 6. Apply data-quality, observability, and governance controls across pipelines 7. Build a working Mini-Lakehouse with Delta Lake and PySpark 8. Connect Lakehouse components to AI, reporting, and analytics consumers
Topics	1. Evolution of data architecture: warehouse, lake, Lakehouse 2. Compute-storage separation and cloud economics 3. ETL versus ELT in modern stacks 4. Delta Lake and ACID transactions 5. Streaming data and event-driven architectures 6. Data quality, observability, and governance 7. Hands-on: building a Mini-Lakehouse 8. Serving data to AI and BI workloads
Tools & Platforms	PySpark, Delta Lake, dbt, Kafka basics, Great Expectations
Assessment	Labs; Mini-Lakehouse build project
Stackability	Data-engineering badge; anchors the Data Engineering for AI specialisation. Derived from the existing Modern Data Engineering programme. Next: SDA-DSC-215 / SDA-DSC-313

SDA-DSC-215 — Big Data Analytics with Spark

Spark تحليلات البيانات الضخمة باستخدام

Level / المستوى	Specialist / متخصص
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	A specialist module on analysing datasets too large for a single machine. Participants master distributed computing concepts, write performant Spark transformations and SQL, and tune jobs through partitioning and caching. The module scales every prior analytics skill to enterprise data volumes.
الوصف	وحدة تخصصية لتحليل مجموعات البيانات التي تتجاوز قدرة جهاز واحد. يتقن المشاركون مفاهيم عالية الأداء وضبط المهام عبر التقسيم والتخزين SQL و Spark الحوسبة الموزعة وكتابة تحويلات المؤقت، بما يوسع كل مهارات التحليل السابقة إلى أحجام البيانات المؤسسية.
Target Audience	Data scientists and engineers working with large-scale data
Prerequisites	SDA-DSC-214 or equivalent Spark exposure
Learning Outcomes	1. Explain distributed execution: partitions, shuffles, and lazy evaluation 2. Develop analytical workloads with Spark DataFrames and Spark SQL 3. Optimize job performance through partitioning, caching, and join strategies 4. Implement scalable feature-engineering and ML pipelines with Spark MLlib 5. Analyze query plans to diagnose and resolve performance bottlenecks 6. Design batch analytics workflows on Lakehouse storage
Topics	1. Distributed computing fundamentals 2. Spark DataFrames and Spark SQL 3. Shuffles, partitioning, and caching 4. Performance tuning and query plans 5. Spark MLlib pipelines at scale 6. Big-data analytics project
Tools & Platforms	PySpark, Spark SQL, MLlib, Delta Lake
Assessment	Labs; large-scale analytics project
Stackability	Big-data badge; part of the Data Engineering for AI specialisation. Next: SDA-DSC-313

SDA-DSC-311 — Decision Science and Optimization Modeling

علم القرار ونمذجة التحسين

Level / المستوى	Expert / خبير
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	An expert module that moves from predicting outcomes to prescribing actions. Participants formulate linear and integer optimisation models, run simulations under uncertainty, and combine ML predictions with optimisation to recommend optimal decisions. Applications span resource allocation, scheduling, and policy planning.
الوصف	وحدة خبراء تنتقل من التنبؤ بالنتائج إلى وصف الإجراءات. يصوغ المشاركون نماذج التحسين الخطي والصحيح وينفذون المحاكاة في ظل عدم اليقين ويجمعون تنبؤات تعلم الآلة مع التحسين للتوصية بالقرارات المثلى، مع تطبيقات في تخصيص الموارد والجدولة وتخطيط السياسات.
Target Audience	Senior data scientists supporting operational and strategic decisions
Prerequisites	SDA-DSC-211; SDA-DSC-213 recommended
Learning Outcomes	1. Formulate business problems as linear and integer optimisation models 2. Implement optimisation solutions using modern solver frameworks 3. Design simulations to evaluate decisions under uncertainty 4. Integrate ML predictions into prescriptive decision pipelines 5. Analyze trade-offs with sensitivity and scenario analysis 6. Evaluate decision-support systems for robustness and adoption
Topics	1. From predictive to prescriptive analytics 2. Linear and integer programming 3. Solver frameworks and model building 4. Monte Carlo simulation 5. Predict-then-optimise pipelines 6. Sensitivity and scenario analysis 7. Decision-science case project
Tools & Platforms	PuLP/OR-Tools, SimPy, pandas
Assessment	Optimisation project with decision memo
Stackability	Decision-science badge; Expert elective. Next: SDA-DSC-390

SDA-DSC-312 — GenAI-Augmented Data Science Workflows

سير عمل علم البيانات المعزز بالذكاء التوليدي

Level / المستوى	Expert / خبير
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	An expert module on multiplying data-science productivity with generative AI. Participants use LLMs for code generation, automated EDA, synthetic data, and unstructured-text analysis, while building reproducible, verifiable workflows. The module defines the professional standard for responsible AI-assisted analysis.
الوصف	وحدة خبراء لمضاعفة إنتاجية علم البيانات بالذكاء التوليدي. يستخدم المشاركون النماذج اللغوية لتوليد الكود والتحليل الاستكشافي الآلي والبيانات الاصطناعية وتحليل النصوص غير المهيكلة مع بناء سير عمل قابل للتكرار والتحقق، وتحدد الوحدة المعيار المهني للتحليل المدعوم بالذكاء الاصطناعي.
Target Audience	Experienced data scientists modernising their workflow
Prerequisites	SDA-DSC-211; SDA-FND-102
Learning Outcomes	1. Integrate AI coding assistants into analysis workflows with verification discipline 2. Apply LLMs to extract structure and insight from unstructured text data 3. Develop synthetic-data strategies for augmentation and privacy protection 4. Design reproducible, version-controlled analysis pipelines with AI assistance 5. Evaluate AI-generated analytical artefacts for correctness and bias 6. Implement agent-assisted automation for recurring analytics tasks
Topics	1. AI-assisted coding and analysis patterns 2. LLMs for text analytics and feature extraction 3. Synthetic data generation and privacy 4. Reproducibility and verification discipline 5. Automating recurring analytics with agents 6. Responsible use and quality assurance
Tools & Platforms	AI coding assistants, LLM APIs, Git, notebook-to-pipeline tooling
Assessment	Augmented-workflow project with verification report
Stackability	GenAI-workflow badge; Expert elective bridging to AI Engineer track. Next: SDA-DSC-390

SDA-DSC-313 — Real-Time and Streaming Data Analytics

التحليلات اللحظية وتدفقات البيانات

Level / المستوى	Expert / خبير
Duration / المدة	3 days × 5 learning hours = 15 hours
Description	An expert module on analytics that cannot wait for batch jobs. Participants build streaming pipelines with Kafka and Spark Structured Streaming, implement windowed aggregations and real-time feature computation, and serve live metrics and alerts. The module completes the modern data stack for real-time AI applications.
الوصف	وحدة خبراء في التحليلات التي لا تحتمل انتظار المعالجة الدفعية. يبني المشاركون خطوط تدفق باستخدام Kafka و Spark Structured Streaming وخصائص التجميعات النافذة وحساب الخصائص اللحظية ويقدمون مقاييس وتنبيهات حية، بما يكمل منظومة البيانات الحديثة لتطبيقات الذكاء الاصطناعي اللحظية.
Target Audience	Senior data scientists and engineers building real-time products
Prerequisites	SDA-DSC-214; SDA-DSC-215 recommended
Learning Outcomes	1. Design streaming architectures with appropriate delivery and consistency guarantees 2. Implement event ingestion and processing with Kafka and Structured Streaming 3. Develop windowed aggregations, joins, and stateful stream computations 4. Build real-time feature pipelines serving online ML models 5. Implement live dashboards, alerting, and anomaly detection on streams 6. Evaluate latency, throughput, and correctness trade-offs in streaming systems
Topics	1. Streaming architecture and event-driven design 2. Kafka fundamentals and event modelling 3. Spark Structured Streaming 4. Windowing, watermarks, and state 5. Real-time features for online ML 6. Live monitoring and anomaly detection 7. Streaming analytics project
Tools & Platforms	Kafka, Spark Structured Streaming, Delta Live patterns, Grafana
Assessment	Streaming pipeline project
Stackability	Streaming badge; completes the Data Engineering for AI specialisation. Next: SDA-DSC-390

SDA-DSC-390 — Data Science Capstone: From Data to Decision

مشروع التخرج: من البيانات إلى القرار

Level / المستوى	Expert / خبير
Duration / المدة	5 days × 5 learning hours = 25 hours
Description	The integrative capstone of the Data Scientist pathway. Teams take an ambiguous organisational question through framing, data engineering, modelling, validation, and a decision-ready recommendation, defended before an expert panel. The project produces a portfolio-grade artefact demonstrating end-to-end professional competence.
الوصف	المشروع التكاملي الختامي لمسار عالم البيانات. تأخذ الفرق سؤالاً مؤسسياً غامضاً عبر الصياغة وهندسة البيانات والنمذجة والتحقق وصولاً إلى توصية جاهزة للقرار تُعرض أمام لجنة خبراء، وينتج المشروع عملاً احترافياً يثبت الكفاءة المهنية المتكاملة.
Target Audience	Candidates completing the Data Scientist Expert certificate
Prerequisites	SDA-DSC-211 + SDA-DSC-213 + SDA-DSC-214 (or equivalent portfolio)
Learning Outcomes	1. Frame an ambiguous organisational question as a tractable analytical problem 2. Design and build the data pipeline supporting the analysis 3. Develop and validate models with rigorous, leakage-free methodology 4. Analyze results for robustness, fairness, and practical significance 5. Construct a decision-ready recommendation with quantified uncertainty 6. Defend methodology and findings before a technical panel
Topics	1. Problem framing and scoping workshop 2. Data acquisition and pipeline build 3. Modelling and validation sprints 4. Robustness, fairness, and sensitivity checks 5. Decision memo and visual narrative 6. Final presentation and panel defence
Tools & Platforms	Full track toolchain as selected by teams
Assessment	Portfolio project; decision memo; panel defence
Stackability	Completes the Data Scientist Expert (Professional) Certificate.

8. Scalability and Future Expansion

- New role tracks (AI Product Manager, AI Solution Architect, ML Researcher) plug into the same Foundation Core and coding scheme.
- Modules are versioned independently; fast-moving content (agentic AI, LLMOps) can be refreshed without touching the pathway structure.
- Cross-track electives: Expert learners may take one module from the sister track for certificate credit, reinforcing T-shaped profiles.
- Delivery formats (in-person, hybrid, cohort-online) are decoupled from module definitions to enable regional scaling.